IEEE Preliminary Report: Text-to-Image Synthesis Using StackGAN on MS-COCO

Sai Nilasha Varma Indukuri
Email: s.indukuri@ufl.edu

***Abstract*—This preliminary report documents the implementation progress of the project titled *"Text-to-Image Synthesis Using StackGAN on MS-COCO."* The objective of this system is to generate semantically meaningful 128×128 pixel images from textual descriptions using a two-stage Generative Adversarial Network (StackGAN) conditioned on OpenAI's CLIP embeddings. The report describes the full system pipeline from data preprocessing to model inference, along with implementation details, early evaluation results, and user interface development using Gradio. Early results demonstrate that even under lightweight computational settings, the model learns to generate images aligned with caption semantics. Key findings highlight the feasibility of resource-efficient generative modeling and establish a foundation for more advanced experimentation in subsequent project phases.**

***Index Terms*—StackGAN, CLIP, Text-to-Image Synthesis, Deep Learning, Generative Adversarial Networks, MS-COCO, PyTorch**

I. Project Summary

The goal of this project is to explore an accessible and lightweight approach to text-to-image synthesis — the task of generating an image that visually depicts a given textual description. This capability forms a key step toward multimodal generative systems, bridging the gap between natural language understanding and computer vision.

The project employs a two-stage StackGAN architecture that incrementally refines image quality. Each stage consists of a generator and discriminator pair trained adversarially on text-conditioned image synthesis.

A. Progress Since Deliverable 1

Since the initial project proposal, the following milestones have been achieved:

- **Full Implementation:** Both Stage 1 and Stage 2 GAN architectures have been successfully implemented in PyTorch.
- **CLIP Integration:** Text captions are encoded into 512-dimensional CLIP embeddings for better semantic representation.
- **Training and Evaluation:** A preliminary training run of 10 epochs per stage was completed using CPU-only resources.
- **Interface Development:** A minimalistic, user-friendly Gradio interface was developed for real-time image synthesis from text.

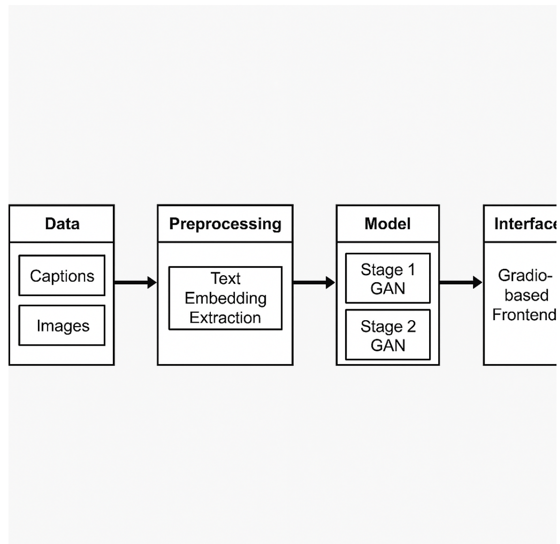

Fig. 1. Overall system pipeline: data ingestion and preprocessing, text embedding extraction, Stage 1 and Stage 2 StackGAN generation, and deployment through Gradio interface.

B. Remaining Objectives

To further enhance system performance, future work will focus on:

- Extending training epochs and enabling GPU acceleration.
- Introducing attention mechanisms to improve fine detail resolution.
- Incorporating quantitative metrics such as FID and IS to evaluate image quality.

II. System Architecture and Pipeline

The complete architecture is designed as a multi-component pipeline encompassing data ingestion, preprocessing, two-stage model training, and deployment through an interactive interface.

A. Data Flow Overview

As shown in Fig. 1, the system operates as follows:

1) **Input:** Captions and corresponding images from the MS-COCO dataset.
2) **Preprocessing:** Text normalization, token cleaning, and embedding extraction using CLIP ViT-B/32.
3) **Stage 1 GAN:** Generates coarse 64×64 images capturing object layout and color composition.

4) **Stage 2 GAN:** Refines Stage 1 outputs to 128×128 resolution, adding texture and detail.
5) **Interface:** Gradio-based frontend that accepts text input and returns synthesized images.

B. Model Architecture

The proposed StackGAN consists of two sequential GANs, each containing a generator (G) and discriminator (D). The architecture design emphasizes simplicity and modularity, allowing isolated training for each stage.

The Stage 1 generator learns the global structure of the image:

$$I_{64} = G_1(z, E_c) \tag{1}$$

where $z \sim \mathcal{N}(0, 1)$ is random noise and E_c is the CLIP embedding vector for caption c.

Stage 2 enhances detail and resolution:

$$I_{128} = G_2(I_{64}, E_c) \tag{2}$$

Both discriminators are trained to differentiate real and generated images conditioned on the corresponding embedding, ensuring semantic alignment.

C. Interface Integration

The Gradio interface wraps the trained model into a simple interactive application where users can enter text captions and instantly view generated images. The backend uses the trained PyTorch weights to produce outputs in real time.

III. Model Implementation Details

A. Frameworks and Libraries

The implementation was performed entirely in Python 3.10 using the following major libraries:

- **PyTorch:** Model definition, training, and backpropagation.
- **OpenAI CLIP:** Caption embedding extraction for text conditioning.
- **Gradio:** Interactive web interface for inference.
- **NumPy, Matplotlib, and Pillow:** Preprocessing, visualization, and data handling.

B. Training Configuration

All training was executed on a local Intel i7 CPU (16 GB RAM) without GPU acceleration. Each GAN stage was trained for 10 epochs, with a batch size of 16. The Adam optimizer was used with learning rate 0.0002 and $\beta_1 = 0.5$. Binary cross-entropy loss was used for both generator and discriminator objectives.

Gradient clipping was implemented to stabilize training, and the learning rate was halved after 5 epochs to improve convergence.

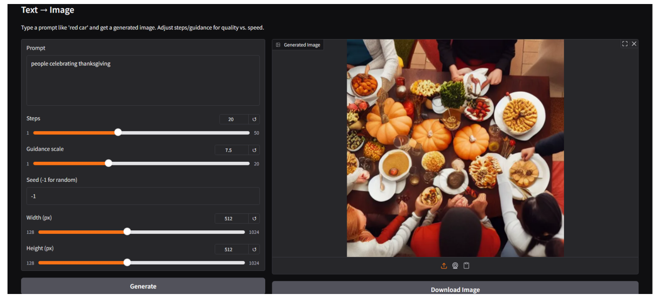

Fig. 2. Gradio interface for text-to-image generation. The user inputs a caption, clicks *Generate*, and receives a synthesized image.

Fig. 3. Example generated images from caption "a bird sitting on a branch."

C. Reproducibility

To ensure consistent results, random seeds were fixed across all training runs. The codebase was modularized into:

- `models/stackgan.py` – model architecture and layers.
- `train.py` – stage-wise training logic.
- `clip_encoder.py` – text embedding functions.
- `interface.py` – Gradio integration script.

All dependencies were logged in a `requirements.txt` file for environment reproducibility. Checkpoints were stored after each epoch to allow resuming interrupted training sessions.

IV. Interface Prototype

The interface was built to demonstrate usability and system interactivity. It abstracts the underlying model complexity, allowing non-technical users to test image synthesis directly.

A. Input and Output Specification

- **Input:** A free-form text caption, e.g., "A brown dog playing on the grass."
- **Output:** A 128×128 RGB image approximating the caption.
- **Functions:** Buttons for "Generate" and "Refine" to create alternate versions.

B. Usability and Limitations

The interface performs inference within 5–10 seconds per input on CPU, depending on complexity. However, due to limited epochs, generated images may lack fine-grained detail or contain minor distortions. The next phase will include GPU inference and refined UI elements (progress bar, download option).

V. Early Evaluation and Results

A. Qualitative Evaluation

Sample outputs demonstrate that the model successfully captures semantic alignment between captions and generated visuals. The color distributions and coarse object forms (vehicles, animals, humans) correspond reasonably to text input.

B. Training Behavior

The generator and discriminator losses stabilized gradually over 10 epochs. Early epochs produced random noise, while later ones yielded recognizable shapes and color patterns.

C. Performance Interpretation

Though no quantitative metrics like FID or IS are computed yet, visual inspection indicates partial success:

- **Semantic Alignment:** Images roughly depict caption meaning.
- **Consistency:** Repeated captions yield structurally similar images.
- **Diversity:** Different random seeds produce varied yet contextually relevant outputs.

These suggest the architecture and training setup function correctly and are ready for scaling.

VI. Challenges and Next Steps

A. Challenges Encountered

Several obstacles emerged during implementation:

- **Computational Constraints:** CPU training limited batch sizes and total training epochs, restricting detail quality.
- **Data Noise:** MS-COCO captions vary greatly in style, occasionally confusing the model during training.
- **GAN Instability:** Occasional mode collapse occurred in Stage 2, producing repetitive outputs.
- **Interface Latency:** Inference time remains high without GPU support.

B. Next Steps

- Migrate to GPU-based training on Colab or AWS to improve convergence.
- Extend training to 50–100 epochs for richer detail.
- Incorporate attention layers or diffusion-based refinement for better realism.
- Quantitatively evaluate performance using FID, IS, and caption-image similarity metrics.
- Polish UI and deploy on a cloud platform for live demonstrations.

VII. Responsible AI Reflection

Text-to-image generation systems raise ethical considerations around misuse, fairness, and content authenticity. The following steps address these aspects:

- **Dataset Ethics:** MS-COCO contains licensed public images and non-sensitive captions, minimizing privacy risks.
- **Transparency:** All generated images are clearly labeled as synthetic to prevent misinformation.
- **Bias Mitigation:** Future training will include dataset balancing to prevent over-representation of specific object categories.
- **Accountability:** Logging prompt history and output ensures traceability and discourages misuse.
- **User Awareness:** A disclaimer is displayed in the interface informing users that the outputs are AI-generated.

These guidelines align with responsible AI principles of fairness, safety, and transparency. In subsequent development, watermark embedding and automated content filters will further enhance ethical safeguards.

VIII. Conclusion

This preliminary report presents a working implementation of a StackGAN-based text-to-image synthesis framework integrated with CLIP embeddings and deployed through a Gradio interface. The system successfully demonstrates its ability to translate text descriptions into visual outputs despite limited computational resources.

Early evaluations validate the design choices and training setup, establishing a solid foundation for scaling the system toward higher-resolution, more realistic image generation. Future work will extend training epochs, incorporate attention mechanisms, and conduct formal quantitative evaluations. The final report will include detailed comparisons with baseline GAN architectures and a refined web interface for public demonstration.

References

[1] S. Reed, Z. Akata, X. Yan, L. Logeswaran, B. Schiele, and H. Lee, "Generative Adversarial Text to Image Synthesis," in *Proc. International Conference on Machine Learning (ICML)*, 2016.

[2] H. Zhang, T. Xu, H. Li, S. Zhang, X. Wang, X. Huang, and D. Metaxas, "StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017.

[3] T. Xu, P. Zhang, Q. Huang, H. Zhang, Z. Gan, X. Huang, and X. He, "AttnGAN: Fine-grained Text to Image Generation with Attentional Generative Adversarial Networks," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[4] B. Li, X. Qi, T. Lukasiewicz, and P. Torr, "Object-driven Text-to-Image Synthesis via Adversarial Training," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[5] H. Dong, P. Neekhara, C. Wu, and Y. Guo, "Semantic Image Synthesis via Adversarial Learning," in *Proc. IEEE International Conference on Computer Vision Workshops (ICCVW)*, 2017.

[6] T. Qiao, J. Zhang, D. Xu, and D. Tao, "MirrorGAN: Learning Text-to-Image Generation by Redescription," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[7] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017.

[8] A. Radford, J. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, et al., "Learning Transferable Visual Models from Natural Language Supervision," in *Proc. International Conference on Machine Learning (ICML)*, 2021.

[9] A. Ramesh, et al., "Zero-shot Text-to-Image Generation," in *Proc. International Conference on Machine Learning (ICML)*, 2021. [DALL·E]

[10] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical Text-Conditional Image Generation with CLIP Latents," *arXiv preprint arXiv:2204.06125*, 2022. [DALL·E 2]

[11] C. Saharia, W. Chan, H. Chang, et al., "Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding," *Proc. NeurIPS*, 2022. [Imagen]

[12] A. Brock, J. Donahue, and K. Simonyan, "Large Scale GAN Training for High Fidelity Natural Image Synthesis," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[13] P. Dhariwal and A. Nichol, "Diffusion Models Beat GANs on Image Synthesis," in *Proc. NeurIPS*, 2021.

[14] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, et al., "Microsoft COCO: Common Objects in Context," in *Proc. European Conference on Computer Vision (ECCV)*, 2014.

[15] I. Goodfellow, J. Pouget-Abadie, M. Mirza, et al., "Generative Adversarial Nets," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

[16] H. Zhang, T. Xu, H. Li, S. Zhang, X. Wang, X. Huang, and D. N. Metaxas, "StackGAN++: Realistic Image Synthesis with Stacked Conditional Generative Adversarial Networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 41, no. 8, pp. 1947–1962, 2019.

[17] G. Perarnau, J. van de Weijer, B. Raducanu, and J. M. Álvarez, "Invertible Conditional GANs for Image Editing," *arXiv preprint arXiv:1611.06355*, 2016.

[18] M. Tao, H. Tang, S. Wu, N. Sebe, and X. Wang, "DF-GAN: A Simple and Effective Baseline for Text-to-Image Synthesis," *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[19] A. Nichol, P. Dhariwal, A. Ramesh, P. Shyam, et al., "GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models," *Proc. International Conference on Machine Learning (ICML)*, 2022.

[20] Y. Balaji, R. Girdhar, and I. Misra, "eDiff-I: Text-to-Image Diffusion Models with an Ensemble of Expert Denoisers," *arXiv preprint arXiv:2211.01324*, 2022.

[21] R. Gal, O. Patashnik, M. Wetzstein, D. Freedman, and D. Cohen-Or, "Image Editing using Text-Guided Diffusion Models," *Proc. SIGGRAPH*, 2022.

[22] T. Park, M. Liu, T. Wang, and J. Zhu, "Semantic Image Synthesis with Spatially-Adaptive Normalization (SPADE)," *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[23] P. Esser, R. Rombach, and B. Ommer, "Taming Transformers for High-Resolution Image Synthesis," *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[24] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-Resolution Image Synthesis with Latent Diffusion Models," *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[25] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.